

Enhancing Emergency Medication Decisions with Machine Learning based Drug Recommendation System

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Abstract: In healthcare facilities worldwide, staff shortages and equipment limitations often lead to overburdened systems, reducing the availability of proper treatment and resulting in numerous avoidable deaths. Self-medication, without expert guidance, exacerbates the situation as individuals tend to worsen their conditions using unsafe methods. Machine learning applications have revolutionized various industries, leading to a surge in automation development. This study introduces a medical prescription system designed to ease the workload of healthcare professionals. The system analyzes patient feedback to predict sentiment by using several vectorization techniques, progressing to Term Frequency-Inverse Document Frequency (TFIDF), and finally incorporating Manual Feature Analysis. A classification algorithm within the system selects appropriate medications based on different medical conditions. The effectiveness of the sentiment prediction is evaluated using precision, recall, F1 score, and accuracy metrics to measure its performance. This system aims to improve healthcare efficiency and patient outcomes by automating the prescription process.

I. INTRODUCTION

The COVID-19 pandemic has created unprecedented demands for healthcare providers, pushing security operations to historically high levels across the globe. During this time, medical professionals are facing shortages, and essential medical supplies are becoming increasingly scarce. This has led to a situation where the demand for healthcare far exceeds the available resources, resulting in significant barriers to effective healthcare management. Due to the inability to doctors, many people resort to self-medication, improper treatments without professional evaluation. In recent years, machine learning (ML) has proven to be a transformative tool in various industries, including healthcare. By utilizing ML technology, healthcare providers can make more informed decisions and predict disease outcomes through automated protocols that help alleviate some of the challenges in patient care. Drug recommendation systems are particularly promising, as they assist healthcare professionals in selecting the most appropriate medications based on the specific needs of patients.

A. Objective of the study

The primary focus of this research was a computerized system which will be based on machine learning algorithms and provide recommendations for medicines based on disease classifications. This algorithm uses feedback data and sentiment analysis to make patient-targeted drug recommendations for health-care providers. The study makes use of various vectorization methods such as TFIDF.[1] The inference was that using sentiment analysis approaches invariably led to a better prediction performance when put together within the model. Thus, two primary benefits: The platform helps with crisis management for medical staff and lessens treatment determination turnaround time due to nominations made by the model.

B. Scope of the study

The present research is designed to provide an intelligent recommendation for a drug. This combines Machine learning systems with sentiment analytical features.

It covers using different vectorization techniques for processing textual data; a sentiment analysis approach designed for all nursing homes to evaluate feedback reviews. The research then uses many different classification algorithms to produce the most beneficial drugs inducted as medications. The evaluation metrics are calculated through precision and recall, combined with accuracy and F1 score and AUC score. An analysis of text based patient feedback relationships with drug recommendations, while also striving to advance system performance tools to assist medical staff.

C. Problem Statement

Several shortages have arisen in the healthcare sector due to the coronavirus pandemic. In combination with shortages of medical staff, lack of equipment and medicine has led to further demise of countless people. Most people lack proper medical directions, so they resort to self-medication, leading to deteriorating health conditions. Modern healthcare systems reap the advantages of effective machine learning methods that drive active research towards automated healthcare development.[2]

The current system presents a medicine recommendation platform that aims to support overloaded healthcare staff who struggle to go through certain procedures. Our system analyzes the patient review along with the sentiment prediction by employing text-processing attributes like TF-IDF and Manual Feature Analysis to recommend a specific medicine. When the model shared results, only the Random Forest classifier with TF-IDF vectorization achieved the best accuracy rates for predictive sentiment. The evaluation metrics were: precision, recall, F1 score, accuracy, AUC score.

II. RELATED WORK

Methods of machine learning have been the focus of this research since they analyze drug feedback while extracting public sentiment about medicines. This research demonstrates how sentiment analysis achieves satisfactorily in classifying drug reviews into positive and negative assessments and neutral feedback in order to create base knowledge for recommendation systems. Medical providers gain detection capability over trends in sentiment, which results in the ability to convert customer feedback into targeted drug recommendations, raising the quality of the recommendations to effective public-sentiment-preferred medical treatments. [3] The research focuses on the development of a market dominating drug recommending platform on emul-

ating sentiment analysis to medicine effect predictions and outcome measures in terms of patient satisfaction. The reviews from users will be used in the creation of drug oriented sentiment categories, leading to prescription recommendations in the context of the starting patients. Sentiment status evaluation is, indeed, of immense importance in bespoke medication choices, for literature exists providing enough evidence that sentiment-based algorithms produce better outcomes in therapy recommendations.[4]

The assessment reviews various approaches of sentiment analysis applied in drug review evaluation systems for their effectiveness in both positive and negative assessments.

Sentiment analysis supported by either NN or SVM approach improves the prediction accuracy for drug satisfaction levels. Results support a complete matching solution for a hybrid of sentiment analysis with drug recommendation systems by merging medical criteria with consumer reviews. [5]

The work improves pharmaceutical recommendation algorithms by merging collaborative filtering methods with sentiment analysis assessment approaches. The system analyzes user preferences with sentiment indicators from posts to suggest medicine based on both profile compatibility and the user's sentiment history. The combination of collaborative filtering and sentiment analysis generates optimum customized drug suggestions yielding greater patient satisfaction and better medical outcomes. [6]

Research aims to improve drug recommendation systems with sentiment feedback analysis techniques. The model scans user feedback across platforms related to medicine for patterns of satisfaction, which are used to make medication recommendations on the best-rated drugs. The analysis of research findings indicates that sentiment-based feedback analytics provide a dramatic improvement in recommendation systems. Further, it creates a very adaptive resource that enhances both the patients and the healthcare providers' abilities in selecting medications. Such recommendations with very supportive material help the doctors know that choosing patients' emotional feedback together with medical information yields better decisions. [7]

III. PROPOSED SYSTEM

The proposed framework operates to improve the accuracy of medicine sentiment classification through a solution against diagnostic inconsistencies and data

C. EDA (Exploratory Data Analysis)

[illegible]

Figure 1 Word Cloud

The prominence of the word "side" indicates that side effects are a serious worry. Menstrual cycle and timing issues are often mentioned, with "period" and "time" being important terms. Words like "take" and "pill," which refer to the act of taking medication, are frequently discussed. Emphasizing the length of time and frequency of medication use are terms like "day" and "month." The importance of medical personnel and pharmaceuticals is demonstrated by the popularity of the terms "doctor" and "medication." In the conversations, common symptoms or sensations like pain, worry, and feeling ill stand out.

D. Model Building

The stated goal is to use the attributes drugName, condition, usefulCount, and rated sentiment to create a model structure predicting drug review sentiment. The classification problem requires execution through Decision Trees and Random Forests as well as XGBoost and Logistic Regression models. The models inspect variable attribute relationships to establish whether reviews express neutral or negative or favorable sentiment. [10] After training the models on training and testing data sets, the assessment utilizes accuracy, precision, and recall to measure their performance. Further tuning could yield an enhancement in models' performance levels through hyperparameter adjustment, so as to train the most efficient model that can then predict the tone of new reviews, hence giving useful insights into customer medication feedback.

This requires an analysis of accuracy, coupled with precision, recall, and F1-score. Accuracy, which is the total number of successful predictions divided by the number of predictions made, can be used to evaluate the overall performance of the model. Precision would highlight the imbalanced class distributions since it gives

a balanced measure of recall and precision. Visually, these numbers indicate how well the model identifies review sentiment.

DRUG RECOMMENDATION SYSTEM

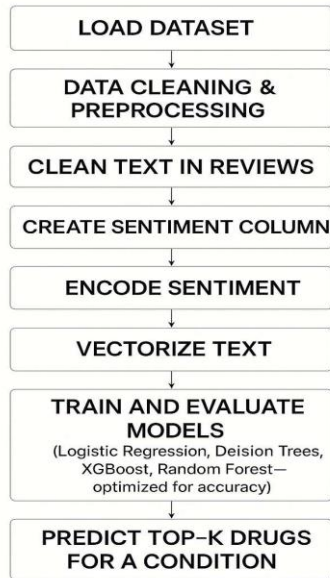


Figure 2 Block diagram for proposed system

IV. METHODOLOGY

A. Random Forest

Random Forest is an ensemble learning method, building a multitude of decision trees on training data for two purposes: majority voting for classification tasks and averaging predictions for regression tasks. The algorithm randomly selects data subsets from training data and then constructs one decision tree on each subset. With limited random feature selections for each tree, the algorithm counteracts overfitting problems and hence improves generalization ability. When a new observation comes into play, the forest makes predictions based on the collective predictions from all component trees.[11] The distinguishing remark about Random Forests is that they maintain high accuracy throughout while at the same time strong performance with fast execution on both classification problems and regression issues. The modeling framework possesses two unique abilities: : it recognizes important features in data

while it handles missing values.

B. Logistic Regression

The statistical binary classification model Logistic into probabilities for binary outcomes. The logistic model possesses the characteristic of linear associations between independent and dependent elements, applying a sigmoid transformation which maps output values between 0 and 1. It is through this transformation process when the model quithe probability of class assignment. The training of logistic regression determines the predictor weights by finding the optimal solutions using likelihood-based algorithms employing gradient descent during parameter estimation. Model classification decisions rest upon a particularity threshold value of 0.5 for classifying predicting behavior above or below that point. The logistic regression model reveals intelligible predictions that perform generally well when data points populate linearly separable classes.[12]

C. Xgboost Classifier

Extreme Gradient Boosting is thought to be among the best- performing learners today. It is an ensemble learning technique based on decision trees, using the gradient boosting construct. The method builds trees in a sequential manner, creating new trees that attempt to rectify the errors left by the prior trees. The approach employs a form of regularization, together with a penalty for model complexity, to guard against overfitting. Hence, it makes implementing techniques for model generalization a little easier than performing generalization.[13] Using a gradient descent algorithm along with a loss function, such as log loss, XGBoost grows trees in order to minimize error during iterative corrections on the model. XGBoost manages the cases of sparsity through optimized parallel processing algorithms on the basis of enhancing gradient computation; hence it gives faster accuracy than normal gradient boosting models. The whole system is recognized as a star module in the world of machine learning contests and is widely used for classification and regression tasks.

D. Decision Tree

Decision Trees constitute one of the most popular methods in supervised machine-learning classification tasks, which are developed using features carrying the highest information gain or maximum reduction of impurity-determined by Gini impurity or entropy metrics. The construction of the tree starts from the top node, whereby a feature that enables optimal data separation is chosen.[14] Features include maximum depth, minimum samples per

leaf. The organizational system of trees allows for easy interpretation, since any decision is followed through a clear path from the root to a terminating leaf node. Hence, new instances are traversed from the root via decision paths triggered by their feature values, till they reach leaf nodes containing prediction outcomes. Although decision trees are easy to understand, there is a serious risk of being misled, since there is a possibility of overfitting the data unless a well-defined pruning technique is followed.

E. Drug Recommendation:

To recommend the drug for given disease. We use the Drug Review Dataset, which includes disease, drug names, user reviews, ratings, and useful counts.

The drug recommendation has done in 3 steps:

1. Sentiment Analysis & Probability Estimation
2. Weighted Sentiment Score
3. Recommending Top k Drugs

Perform the sentiment analysis on each review using TF-IDF and predict the probability of review to be positive sentiment using Ramdon Forest.

Calculating the weighted sentiment Score:

$$\text{Weighted Sentiment} = P_i \times \left(\alpha \times \frac{R_i}{R_{\max}} + (1 - \alpha) \times \frac{U_i}{U_{\max}} \right)$$

Where α controls the balance between ratings and useful count. Then calculating the Weighted Sentiment Score, we group by drug name and computed mean sentiment score. Finally, drugs are sorted in descending order and top K drugs are recommended.

V. DISCUSSION AND RESULTS

A. Decision Tree

The Model stands exceptionally high because it detects 25,794 true positives alongside 20 omitted positive occurrences. The model indicates a debilitating 16,945 cases of false positives which results in numerous errors when detecting positive cases. The extremely small number of correctly identified actual negatives (15) demonstrates substantial difficulty for the model to detect real negative cases..

The model requires better precision through either threshold adjustment or class balance optimizations to enhance its overall performance.

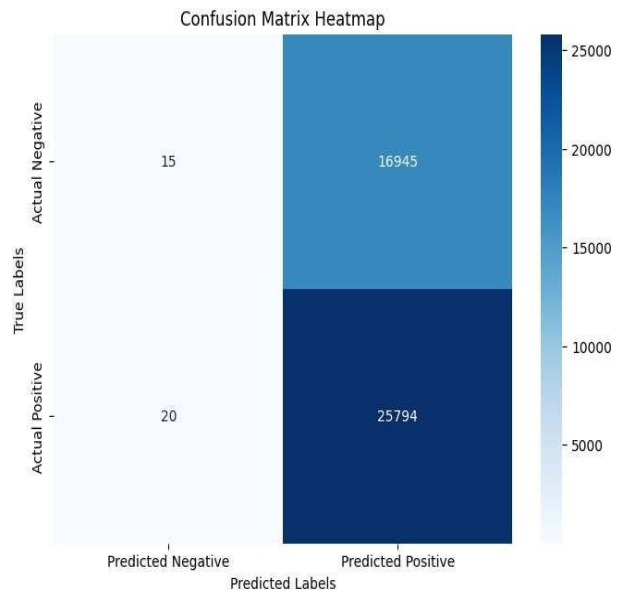


Figure 3 Decision Tree Confusion Matrix

B. Logistic Regression

The model effectively identifies positive cases, with 22,309 true positives. However, it struggles with precision, as indicated by the 4,994 false positives. The high number of false negatives (3,505) suggests the need for improvement in capturing all true positive instances.

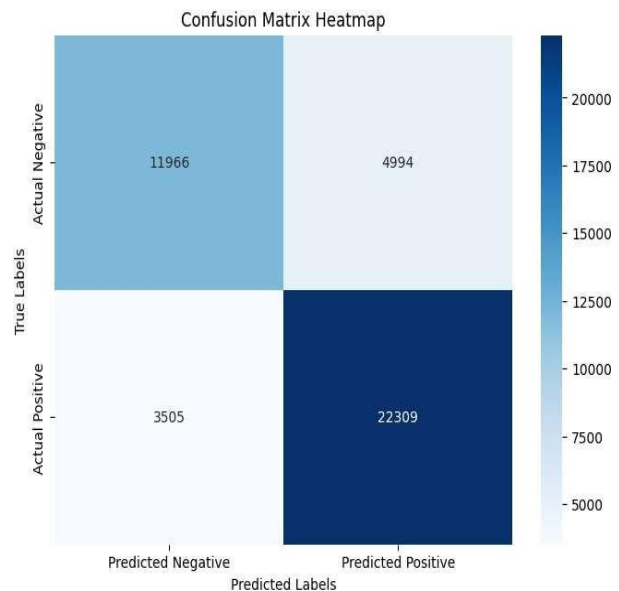


Figure 4 Logistic Regression Confusion Matrix

C. Random Forest

The model accurately identifies true positives (24,702) more often than false positives (2,520), showing strong recognition for actual positive cases. The false negative count (1,112) is relatively low, indicating the model misses fewer true positive instances. However, with 14,440 true negatives and 2,520 false positives, there's room to improve precision in distinguishing between positive and negative classes.

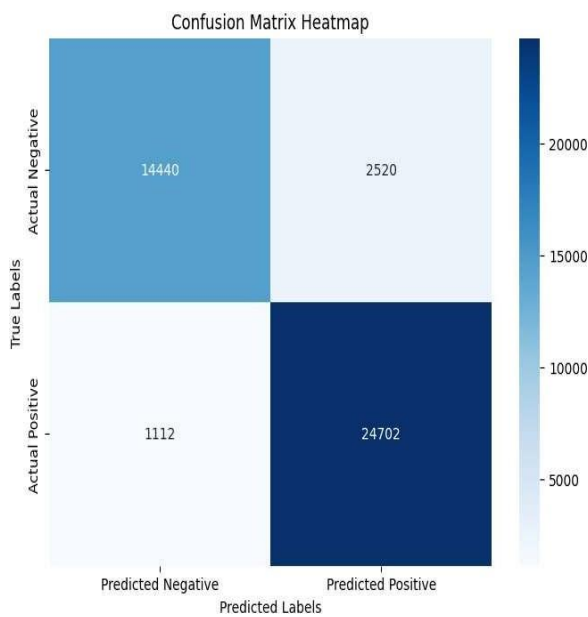


Figure 5 Random Forest Confusion Matrix

D. Xgboost Classifier

The model shows strong performance with 22,838 true positives and 10,983 true negatives, indicating reliable predictions for both classes. The false positive count of 5,977 reveals that the model still struggles with precision in distinguishing positive cases. The presence of 2,976 false negatives suggests there is room for improvement in capturing all true positive instances.

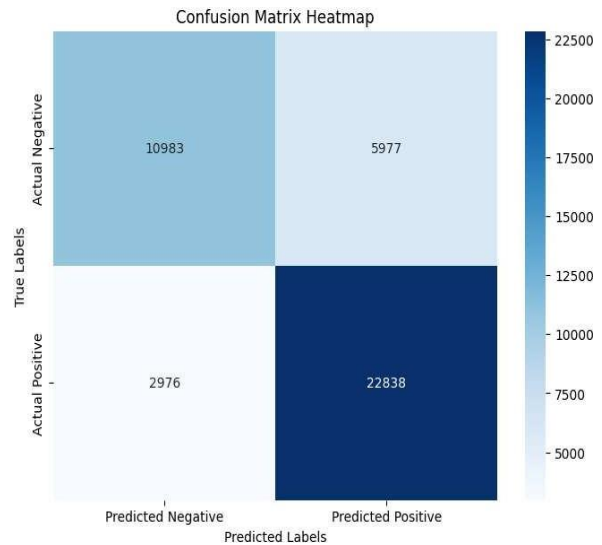


Figure 6 Xgboost Confusion Matrix

E. Accuracy values of each algorithm:

Metric	Logistic Regression	Decision Tree	XGBoost (xgb)	Random Forest (rf)
Accuracy	80%	60%	79%	92%
Precision (Class 0)	77%	43%	79%	93%
Precision (Class 1)	82%	60%	79%	91%
Recall (Class 0)	71%	0%	65%	85%
Recall (Class 1)	86%	100%	88%	96%
F1-Score (Class 0)	0.74	0.00	0.71	0.89
F1-Score (Class 1)	0.84	0.75	0.84	0.93
False Positives (FP)	4,994	16,945	5,977	2,520
False Negatives (FN)	3,505	20	2,976	1,112

Table 1 Accuracy Values



Figure 7 Homepage functionality

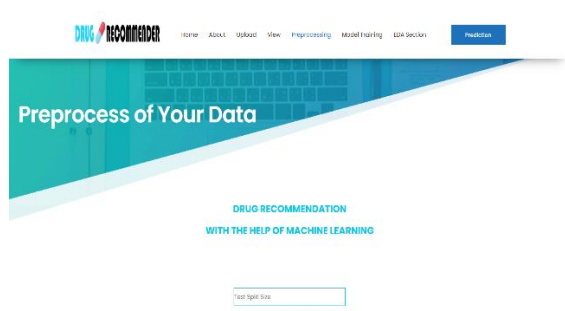


Figure 8 Data Preprocess functionality

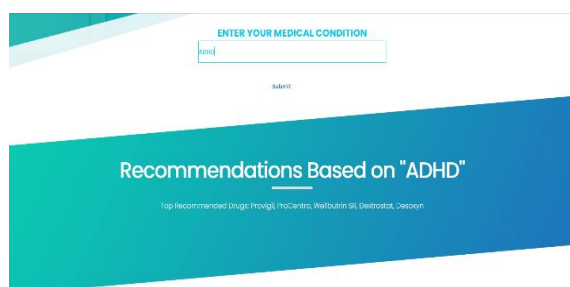


Figure 9 Drug Recommendation functionality

VI. CONCLUSION

Reviews constitute an essential component that people use to make decisions across their daily routines. You can pick the best options by reading reviews when shopping or purchasing online or choosing a restaurant. The increasing popularity of drug review analysis served as the motivation for conducting sentiment analysis in this research. Machine learning classifiers implemented throughout the study included Logistic Regression and Decision Tree and XGBoost and Random Forest with TF-IDF acting as the main feature extraction technique. The classifiers were evaluated using five key metrics: Experiments showed that the Random Forest model operating on TF-IDF features achieved superior accuracy

measures than alternative models. Random Forest established itself as the most accurate model despite the competitors from Decision Tree, LogitRegression, XGBoost.

VII. FUTURE ENHANCEMENT

Moving forward we should extend this project by testing deeper feature extraction methods based on deep learning approaches incorporating Word2Vec or GloVe word embedding systems to enhance textual data handling. We could improve accuracy by developing ensemble methods which use stacking models to combine multiple classifiers.[15] Extending the model's performance scope requires the addition of diverse drug category datasets. A valuable system enhancement would become possible through real-time sentiment analysis which processes reviews immediately after customers post them.

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